

A Unified Theory of Transfers: One Character Framework, a Restricted Classification, and the Number-Field Class-Group Residual

Where the two elementary ECDLP transfers come from, why generic and rational constructions cannot produce a third, and the single evaluator that remains open

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ABSTRACT

The two textbook attacks that break special elliptic curves — the anomalous additive reduction of Semaev, Satoh–Araki and Smart, and the MOV/Frey–Rück pairing reduction — are both instances of one operation: evaluate a nonzero character of a prime-order cyclic source group into a group whose discrete logarithm is easy. We formalise this as a **transfer package**, an injective homomorphism $\varphi_I : \langle P \rangle \rightarrow G_I$ with polynomial setup and evaluation cost and a subexponential target DLP, settle the status of Weil descent inside it (it counts, as a known third geometric family), and ask the problem’s question: does a **structurally different** fourth transfer exist, or can one be ruled out in a stated class? Our answer is a restricted classification with an explicitly isolated residual. We prove (i) an algebraic-group rigidity theorem: every global homomorphism $E \rightarrow H$ into an affine group is trivial and every nontrivial one has elliptic image isogenous to E ; (ii) a generic-source impossibility theorem: any transfer using only a random source-group oracle would, via Shoup’s $O(m^2/r)$ bound, solve the generic DLP in $r^{o(1)}$ queries; (iii) rational-evaluator lower bounds — an affine subgroup homomorphism of common pole degree D forces $D \geq r/2$, a B -branch piecewise map forces $\max(1, D)B^2 \geq r/4$, and the decision-tree depth obeys $d + 2b \geq \log_2 r - O(\log \log r)$; and (iv) a base classification sending finite/local and function-field class targets back to trivial or Jacobian ones. Against these we place the honest boundary. An evaluator-sandwich argument makes point-to-class evaluation polynomial-time equivalent to the source DLP given a target oracle, so class-number size arguments alone cannot close the problem; the surviving candidate is a direct cross-characteristic evaluator into a separately constructed number-field class group, whose discriminant must live in the window $2n - O(\log n) \leq B = o(n^2/\log n)$. We validate both elementary transfers end to end at toy scale, exhibit the exact-order target census, and show why the ray-class shortcut collapses to the source DLP. The result is a defensible restricted classification, not a solution: it names the one object a fourth transfer would have to be.

Keywords. ECDLP · transfer maps · anomalous curves · MOV/Frey–Rück · Weil descent · class groups · generic group model · algebraic groups

1 Introduction

Most of elliptic-curve cryptography rests on a single empirical fact: for a well-chosen curve $E/\mathbb{F}_q$ and a point P of large prime order r , the best known algorithm for the discrete logarithm problem (ECDLP) in $\langle P \rangle$ is a generic square-root method costing $\Theta(\sqrt{r})$ group operations. The qualifier **well-chosen** is doing real work. A handful of curve families are catastrophically weaker, and in each case the break has the same shape: one constructs an efficiently computable, injective group homomorphism from $\langle P \rangle$ into some other group where the discrete logarithm is subexponential or polynomial, evaluates it on the challenge, and finishes in the easy target. Two such **transfers** are classical. Anomalous curves over prime fields, where $\#E(\mathbb{F}_p) = p$, admit the additive reduction of Semaev [1], Satoh–Araki [2] and Smart [3], which maps $\langle P \rangle$ to $(\mathbb{F}_p, +)$ in linear time. Curves with small embedding degree k admit the MOV [4] and Frey–Rück [5] pairing reduction, which maps $\langle P \rangle$ into $\mu_r \subset \mathbb{F}_{q^k}^\times$ and finishes with index calculus in the finite field.

The problem posed as P1.5 asks whether these are the only mechanisms, or merely the only **known** ones. Concretely: **classify the curves for which there is a polynomial-time computable injective homomorphism from $\langle P \rangle$ to a group with a subexponential discrete logarithm algorithm**. This is a question about the existence of transfers, and it is genuinely open in full generality. What we can do — and what this paper does — is give a precise operational definition, identify the common structure of the two known cases, prove that entire natural classes of would-be transfers cannot exist, and isolate the single residual object that a structurally new transfer would have to be. The deliverable is a **restricted classification**, honestly labelled: several branches are closed by proof, one is closed by citation to prior art (Weil descent), and one concrete possibility remains open.

Main finding. The anomalous and pairing transfers are two instances of one construction: a nonzero character of a prime-order source into an easy target, made injective automatically by the prime-order kernel argument. Weil descent is a known third **geometric** family under any honest definition. We prove that no structurally new transfer can be **generic** in the source (it would break Shoup’s bound), **rational of low degree** into an affine target ($D \geq r/2$, and $\max(1, D)B^2 \geq r/4$ with branching), or **rational into a proper or mixed target** (its image is forced to be elliptic and isogenous to E). The only surviving candidate is a direct cross-characteristic evaluator from finite-field point coordinates into a separately constructed number-field class group; an evaluator-sandwich argument shows this cannot be excluded by target size alone, and it remains open.

1.1 Contributions and honest scope

We contribute (i) the transfer-package definition and its boundary examples, with an explicit decision about Weil descent (§3); (ii) a common character framework for the two elementary transfers, anchored in Belding’s dual-number pairing [6] (§3.2); (iii) a candidate-target taxonomy with a reason for every verdict (§4, Figure 1); (iv) the algebraic-group rigidity theorem (§5); (v) the generic-source impossibility theorem (§6); (vi) rational-evaluator degree, branch and depth lower bounds, adversarially audited (§7); (vii) the CM/ray class-group obstructions with explicit class-number bounds (§8); (viii) the evaluator sandwich and discriminant-window analysis that locate the residual (§9); and (ix) end-to-end toy validation of both known transfers and the negative probes (§10, Figure 2, Figure 3, Figure 4).

We state the scope plainly. This is **not** a solution of the unrestricted problem. The classification is complete only within the models it names — global algebraic-group homomorphisms, generic source oracles, rational decision-tree evaluators into affine/proper/mixed targets, and the standard class-target bases. Outside those models, in particular for high-degree or non-rational coordinate/lift/valuation programs into a number-field class group, the question is open, and we record it as such rather than overclaim.

2 Setting and notation

Fix a finite field $\mathbb{F}_q$ of characteristic p and an elliptic curve $E/\mathbb{F}_q$ with identity O . Let $P \in E(\mathbb{F}_q)$ have prime order r and write $C = \langle P \rangle$ for the cyclic source group. An instance is a tuple $I = (q, E, P, r)$ drawn from an infinite family; we set $n = \lceil \log_2 r \rceil$ and let L denote the full encoded input length, requiring $L = n^{O(1)}$ so that “polynomial time” always means polynomial in n . For an imaginary-quadratic order of discriminant Δ we write $h(\Delta)$ for its class number and $B = \log_2 |\Delta|$ for its discriminant bit length. We use mod for reduction, $\langle \cdot \rangle$ for the subgroup generated, and reserve φ for a transfer evaluator. The single standing fact we lean on repeatedly:

Lemma 1 (prime-order kernel). **PROVED** Any nonzero group homomorphism out of a group of prime order is injective, because its kernel is a proper subgroup of a group of prime order and hence trivial.

This is why every transfer question reduces to **nonzero character** plus **easy target**: injectivity is free once the character is nonzero and the source has prime order.

3 The unified transfer framework

We first make “transfer” precise, so that inclusion and exclusion are decidable rather than rhetorical.

3.1 The transfer package

Definition 2 (transfer package). For an infinite family of instances $I = (q, E, P, r)$ with $n = \lceil \log_2 r \rceil$ and encoded input length $L = n^{O(1)}$, a **transfer package** consists of a uniformly and effectively presented finite group G_I , setup data of size polynomial in L , and a uniform evaluator

$$\varphi_I : \langle P \rangle \longrightarrow G_I \quad (1)$$

such that: (1) the encoding length and group-operation cost in G_I are polynomial in L ; (2) $\varphi_I(Q)$ is computable from the ordinary representation of $Q \in \langle P \rangle$ in expected polynomial time in L ; (3) φ_I is a group homomorphism, injective on $\langle P \rangle$; and (4) given $\varphi_I(P)$ and $\varphi_I(Q)$, a uniform target algorithm recovers the logarithm in expected time $\exp(o(n))$, including all precomputation attributable to the instance. The evaluator need not compute φ_I^{-1} ; randomised evaluators and target algorithms are allowed with the same expected-cost bounds.

The four requirements are exactly what separates a genuine attack from a tautology. Requirement (1)–(2) rule out the two most tempting cheats.

Remark 3 (boundary examples). **PROVED** A lookup table of all r source points fails: its setup and representation cost are exponential in n . The abstract isomorphism $[m]P \mapsto m \bmod r$ is **not** a transfer unless its evaluator is supplied independently, because evaluating it **is** the source DLP.

A constant map, the x -coordinate map, and a hash-to-group map fail injectivity, the homomorphism requirement, and the homomorphism requirement respectively. An efficiently computable isogeny to another generic elliptic curve does not qualify merely from being injective: requirement (4) still needs a subexponential target algorithm, which a generic curve does not supply.

3.2 One character framework for the two elementary transfers

Both classical transfers instantiate the same three-step pattern: build an auxiliary structure, evaluate a fixed functional that is a nonzero homomorphism on C , and land in an easy target. Table 1 makes the parallel literal. Belding’s construction [6] of a nondegenerate Weil-pairing analogue on p -torsion over the dual numbers is what turns the loose analogy “both linearise the group” into an actual common mechanism: the anomalous reduction is a pairing attack too, into an additive rather than multiplicative target.

Table 1: The two elementary transfers as one construction (SG-04). Each fixed functional is a nonzero homomorphism on C , hence injective by the prime-order kernel lemma; the auxiliary structures differ but the nondegeneracy certificate is the same. Belding [6] supplies the missing p -torsion pairing that makes the top row literally a pairing attack.

Case	Auxiliary structure	Fixed functional	Easy target	Nondegeneracy
Char. p ($r = p$)	p -adic / dual-number thickening	connecting / formal-log map on p -torsion	$(\mathbb{F}_q, +)$ or additive quotient	image of P nonzero
Prime-to- p	Weil or Tate pairing on $E[r]$	$T \mapsto e_r(T, Q)$, fixed independent Q	$\mu_r \subset \mathbb{F}_{q^k}^\times$	$e_r(P, Q) \neq 1$

CITED Semaev [1] handles all rational characteristic- p torsion, not only the prime-field equation $\#E(\mathbb{F}_p) = p$; the additive family is genuinely broader than the anomalous prompt suggests.

CONJECTURE A genuinely new **elementary** transfer would need a computable nonzero character not induced by a local connecting map, a bilinear torsion pairing, or a geometric descent; any explicit family meeting the transfer definition and provably outside all three refutes the negative assessment.

3.3 The Weil-descent decision

Weil descent is not excluded by terminology; it is admitted when it meets the definition. **CITED** Gaudry–Hess–Smart [7] construct explicit homomorphisms from selected characteristic-two elliptic-curve groups into Jacobians over a smaller field. **PROVED** If a constructed homomorphism has kernel order coprime to r , its restriction to C is injective, since the kernel meets the order- r subgroup trivially; concretely, for a degree- d separable cover $\pi : C' \rightarrow E$ the composite $\pi_*\pi^* = [d]$, so pullback is injective on r -torsion whenever $r \nmid d$. **CONDITIONAL: EGT-2009 family/smoothness hypotheses**

A descent into the low-degree high-genus Jacobian families of Enge–Gaudry–Thomé [8] has a subexponential $L(1/3)$ target algorithm. **PROVED** But merely rewriting $E(\mathbb{F}_{q^n})$ as the rational points of a Weil restriction supplies **no** target algorithm and fails requirement (4). Thus Weil descent is a known **third geometric target family** under the definition — prior art, not a newly discovered mechanism.

4 Taxonomy of candidate targets

The problem asks for a candidate-target table with a reason for every verdict. Figure 1 records ours; §5–§9 supply the proofs. The organising principle is that a target must be able to **contain** an order- r element and to **receive** it homomorphically from the source, and that requirement (4) demands the target DLP be genuinely easier than the source.

	Excluded	Known mechanism	Conditional / neutral	Open (Q004)
Local formal groups	E	K		
Drinfeld-module targets	E			
Tori (global maps / $E[r]$ pairing)	E	K		
Additive char- p ring groups	E			
Endomorphism-order class group	E			
Ray class groups (mod r , r^2)	E			
Jacobians of covers / Weil descent		K	C	
Weil-restriction abelian varieties			C	
Other elliptic curves / AVs	E			
Separately built number-field class group	E			O

■ E — excluded (proved) ■ C — conditional / neutral
■ K — known mechanism (cited) ■ O — open residual

Figure 1: Candidate-target verdicts (SG-05/SG-06/SG-22). E = excluded by proof, K = known established mechanism, C = conditional or neutral, O = open residual. Every additive/affine target is excluded away from $r = p$ by exponent or algebraic-group rigidity; every natural CM/ray class construction is excluded by orientation loss, size, or transparency; covers and descent are the known geometric family; only a separately constructed number-field class group retains an open cell (Q004).

PROVED Local formal groups and additive char- p rings. If r is a unit in the coefficient ring, the formal multiplication series $[r](T) = rT + O(T^2)$ is invertible, so there is no nonzero formal r -torsion; the underlying additive group of a characteristic- p ring is killed by p and has no order- r element for $r \neq p$. The case $r = p$ is the additive/local family of Table 1, not a new target. **CITED Drinfeld modules** have underlying group $\mathbb{G}_a$ [9], so their finite additive points have exponent p ; excluded for $r \neq p$. **PROVED Tori** are affine, so every global algebraic map $E \rightarrow T$ is trivial (§5); the only maps that survive live on $E[r]$ and are the established pairing route. **Jacobians of covers and Weil restrictions** are the geometric branch (§3.3). The subtle rows — the endomorphism-order class group, ray class groups, and a separately constructed number-field class group — are treated in §8–§9.

5 Algebraic-group rigidity

The first structural theorem explains why every **global** attempt to map E into an affine target dies, and why every nontrivial global map stays on the elliptic side.

Theorem 4 (algebraic-group classification). **PROVED** Let k be a finite field, E/k an elliptic curve, H/k a smooth finite-type algebraic group, and $f : E \rightarrow H$ an algebraic-group homomor-

phism. The reduced image of f is either trivial or an elliptic abelian subvariety of H ; in the latter case $f : E \rightarrow f(E)$ is an isogeny. In particular, if H is affine then f is trivial.

Proof. E is proper and connected, so $f(E)$ is a proper connected closed subgroup of H . CITED A complete connected group variety is an abelian variety, and a complete algebraic group is anti-affine [10]. The image has dimension at most one: if zero, connectedness makes it the identity; if one, it is an elliptic curve. In the nonconstant case the kernel is zero-dimensional and finite, so E surjects isogenously onto its elliptic image. If H is affine, the image is both proper connected and affine, so every coordinate function pulls back to a global regular function on the complete curve E , hence to a constant, and f is trivial. $\square$

Remark 5 (why this does not close the problem). PROVED The theorem classifies **global morphisms** $E \rightarrow H$, but a transfer evaluator need only be defined on the finite set $C = \langle P \rangle$ and need not extend to a morphism on E . MOV evades it because fixing the second pairing argument gives a character only on $E[r]$, not a global map $E \rightarrow \mathbb{G}_m$. The anomalous transfer evades it because it uses a lift or infinitesimal thickening and a connecting map, not a global map $E/\mathbb{F}_p \rightarrow \mathbb{G}_a$. The theorem is a rigidity backdrop; the real content is how far a **subgroup-only** evaluator can escape it.

6 Generic-source impossibility

If a would-be transfer touches the source only through the abstract group operations — no coordinates, no lift, no pairing — then it cannot exist. This is the black-box floor.

Theorem 6 (generic-source impossibility). PROVED Fix a uniformly random injective encoding $\sigma : \mathbb{Z}/r\mathbb{Z} \rightarrow S$ and give the algorithm handles $\sigma(1), \sigma(x)$ with only the generic addition, inversion, identity and equality oracles on the source. No infinite family of transfer packages whose setup and evaluation make $r^{o(1)}$ expected source-oracle calls can have both an injective homomorphic evaluator and an $\exp(o(\log r))$ target DLP algorithm.

Proof. Give the alleged package the generic DLP challenge $(\sigma(1), \sigma(x))$. Run its counted setup and compute $h = \varphi(\sigma(1))$ and $y = \varphi(\sigma(x))$. Because φ is a homomorphism, $y = xh$; because the source has prime order and φ is injective, h has order r , so the promised target algorithm on (h, y) returns x . Let $q(r) = r^{o(1)}$ bound the composition's expected source-oracle calls and truncate after $4q(r)/\delta$ calls; by Markov's inequality the truncated generic DLP algorithm still succeeds with probability at least $3\delta/4$, with deterministic query cap $m = r^{o(1)}$. CITED Shoup's random-encoding bound [11] gives success $O(m^2/r) = r^{-1+o(1)}$, which tends to zero — a contradiction. Target representation, group operations and precomputation are harmless: after composition they are arbitrary local computation by the generic source algorithm. $\square$

The theorem draws the exact boundary. CITED The anomalous transfer crosses it through coordinates and a p -adic or dual-number lift; MOV/Frey–Rück crosses it through line functions, an auxiliary torsion point and extension-field arithmetic; Weil descent crosses it through equations, Frobenius, a field basis and divisor maps. PROVED Hence **every** qualifying transfer must exploit a representation-specific source operation — generic ideal, class, ray or level arithmetic on the target side cannot by itself evade the bound. A surviving point-to-class candidate must expose the concrete source operation that extracts scalar-sensitive information; this is a complete black-box classification, not an unrestricted one.

7 Rational-evaluator lower bounds

The next escape is a coordinate formula: a rational map on E , homomorphic only on C , computed from ordinary point coordinates. We measure such a map by its geometric degree and prove it must be either exponentially high-degree, exponentially branched, or linear-depth. Throughout, fix a faithful closed immersion $\rho : H \hookrightarrow \mathrm{GL}_s$ of the affine target and define the **common pole degree** $D = \delta_\rho(F)$ of a rational $F : E \rightsquigarrow H$ as the degree of the least effective divisor $\mathcal{P}(F)$ on $E_{\bar{k}}$ for which every matrix entry of ρF lies in $L(\mathcal{P}(F))$. The base change to $\bar{k}$ is essential: all zero counts happen on the geometric curve.

Theorem 7 (affine rational rigidity). PROVED Let H/k be affine and $F : E \rightsquigarrow H$ be defined at every point of C with $F|_C$ a homomorphism, and put $D = \delta_\rho(F)$. If $r > 2D$ then F is the constant identity. Hence every nontrivial, and so every injective, rational affine transfer satisfies $D \geq r/2$.

Proof. Put $M = \rho(F(P))$. For each entry, $g_{ij}(X) = f_{ij}(X + P) - \sum_\ell f_{i\ell}(X)M_{\ell j}$ has pole degree at most $2D$ (a translate of $\mathcal{P}(F)$ plus $\mathcal{P}(F)$) and vanishes at all r points of C by the subgroup law. Since $r > 2D$, a nonzero function cannot have more zeros than its pole degree, so $g_{ij} \equiv 0$ and $\rho(F(X + P)) = \rho(F(X))M$ identically. Right multiplication by the invertible constant M preserves, at each geometric point, the largest pole order among the entries, so $\mathcal{P}(F)$ is invariant under translation by P . Every translation orbit has r points while $\deg \mathcal{P}(F) = D < r$, forcing $\mathcal{P}(F) = 0$. All entries of ρF are then global regular functions on the proper connected curve E , hence constants; $F(0) = 1$ makes the constant the identity. $\square$

Branching does not rescue the construction, and the improvement over the naive bound is real: the adversarial-partition argument gives a quadratic, not cubic, branch penalty.

Theorem 8 (piecewise overlap obstruction). PROVED Let $\varphi : C \rightarrow H(k)$ be an injective homomorphism agreeing on each of B nonempty branch sets with a rational map of common pole degree at most D , and put $m = \lceil r/B \rceil$ and $D_+ = \max(1, D)$. Then

$$m(m-1) \leq 2D(r-1), \quad \text{hence} \quad D_+ B^2 \geq r/4. \quad (2)$$

Proof. Take a largest branch S , $|S| = \eta \geq m$, with rational map F . For $t \in C$ set $R(t) = |\{x \in S : x + t \in S\}|$; counting ordered pairs of S by difference gives $\sum_{t \neq 0} R(t) = \eta(\eta-1)$. If some $R(t) > 2D$, then at more than $2D$ points both $x, x+t \in S$, so $F(x+t) = \varphi(t)F(X)$; the entrywise defect $\rho(F(X+t)) - \rho(\varphi(t))\rho(F(X))$ has pole degree at most $2D$ and more than $2D$ zeros, hence vanishes identically. As $t \neq 0$ has order r , translation invariance forces $\mathcal{P}(F) = 0$, F constant, and $\rho(\varphi(t)) = I$, contradicting injectivity. Thus every $R(t) \leq 2D$, so $\eta(\eta-1) \leq 2D(r-1)$; monotonicity in $\eta \geq m$ and a short case split on r versus $2B$ give $D_+ B^2 \geq r/4$. $\square$

Corollary 9 (decision-tree depth). PROVED In the rational decision-tree model — base functions of pole degree $\leq D_0$, binary $+, -, \times, /$ gates, at most b binary decisions per path ($B \leq 2^b$ transcripts), arithmetic depth $\leq d$, and M output entries — pole growth gives $D \leq M2^d D_0$, so an injective affine transfer obeys $d + 2b \geq \log_2 r - \log_2(4MD_0)$. For $MD_0 = (\log r)^{O(1)}$ this reads $d + 2b \geq n - O(\log n)$; with no branching, $d \geq \log_2 r - \log_2(2MD_0)$.

CITED These are depth lower bounds in a restricted algebraic model, not superpolynomial-time bounds: depth $\Theta(n)$ with polynomial size is compatible with them, and Miller's pairing program

is exactly that case — a linear-length straight-line program representing a function of expanded degree $\Theta(r)$ [12]. **CITED** The bound also refuses to exclude MOV: a Miller function has divisor of size $\Theta(r)$, so its pole degree already exceeds the $r/2$ threshold, and its polynomial-time evaluation lives entirely in the short-program exit. Proper and mixed targets are equally rigid.

Theorem 10 (proper and mixed targets). **PROVED** (i) A rational $F : E \rightarrow H$ into a smooth proper algebraic group with $F(0) = 1_H$ extends to a global homomorphism $E \rightarrow H^0$, with trivial or elliptic-isogenous image. (ii) Fewer than r rational branches into such an H collapse to one global homomorphism on C , even for disconnected or noncommutative H . (iii) For a single rational map into a smooth algebraic group with Chevalley sequence $1 \rightarrow L \rightarrow H^0 \rightarrow A \rightarrow 1$, the proper projection is global; if the affine cocycle $c(X, Y) = F(X + Y)(F(X)F(Y))^{-1}$ has fibrewise pole degree $D < r$ in each controlled fibre, then F is global with trivial or elliptic-isogenous image.

Proof. **CITED** A rational map from the normal curve E to a proper variety extends uniquely to a morphism [13]; its image lies in the identity component H^0 , an abelian variety over the perfect field k [10], and a zero-preserving morphism of abelian varieties is a homomorphism by rigidity [14]; §5 then classifies the image. For (ii), some branch has distinct X, Y ; setting $t = F(0)$ and $\alpha(Z) = t^{-1}F(Z)$ gives a pointed morphism, and $\beta = \text{Int}(t) \circ \alpha$ is a global homomorphism agreeing with φ on the generator $X - Y$, hence on all of C . For (iii), the projection πF is global by (i), so c is L -valued and vanishes on C^2 ; each entry of $\rho(c) - I$ has the r points of C as zeros and pole degree $D < r$ in each fibre, so it vanishes, F is a rational homomorphism, and the translation trick removes its poles with Galois descent to k [15]. $\square$

The adversarial audit behind these statements — constant maps, translations, disconnected and noncommutative targets, non-faithful representations, exceptional characteristic, and adversarial partitions — is recorded in full in the repository’s rational-transfer review; the closest prior art is interpolation and zero-estimate theory on commutative algebraic groups [16], which does not state the prime-subgroup defect bound, the B^2 tradeoff, or the proper-branch collapse.

8 The CM and ray class-group obstructions

The most natural place to look for a fourth transfer is a class group, where an order- r element is plausible and the Hafner–McCurley relation method [17] gives a subexponential target DLP. Every **natural** class construction nonetheless fails, for three distinct reasons.

8.1 Orientation loss

Lemma 11 (orientation loss). **PROVED** If P has prime order r , then for every $1 \leq m < r$ one has $\langle mP \rangle = \langle P \rangle$ and, for any endomorphism subring R , $\text{Ann}_R(mP) = \text{Ann}_R(P)$, because multiplication by m is an automorphism of the order- r subgroup. Hence any label determined only by the annihilator, the cyclic kernel, or the kernel-isogeny quotient is constant on the nonzero source points and cannot be an injective homomorphism for odd r .

EMPIRICAL: 8 ordinary $j=1728$ curves, $13 \leq p \leq 421$, 368 points Complete enumeration on ordinary $j = 1728$ curves found exactly one CM eigenvalue, annihilator label, cyclic kernel and canonical Vélú quotient per subgroup, matching the lemma (probe_cm_class_targets_full_20260702.csv).

CITED The standard CM construction runs the **other** way — an ideal $\mathfrak{a}$ defines a kernel $E[\mathfrak{a}]$ and an oriented quotient [18] — so it is not a point-to-ideal character.

8.2 The endomorphism class group is too small

Even ignoring orientation, the curve’s own endomorphism class group cannot hold a large-prime image, by an explicit analytic bound.

Proposition 12 (class-number bound). **PROVED** Every imaginary-quadratic order of discriminant Δ satisfies $h(\Delta) \leq (3/\pi)\sqrt{|\Delta|}(\log|\Delta| + 2)^2$. Consequently the endomorphism order of an ordinary $E/\mathbb{F}_q$, whose discriminant divides the Frobenius discriminant $t^2 - 4q$, has $h(\text{End } E) \leq (6/\pi)\sqrt{q}(\log(4q) + 2)^2$, which is below $q/2$ for every $q \geq 2^{21}$. Thus it cannot contain an order- r image when $r \geq q/2$.

Proof. **CITED** The ring class-number formula [19] writes $h(f^2 d_K)$ as $h(d_K)f$ times local factors over a unit index, and the imaginary-quadratic analytic formula [20] gives $h(d_K) = w_K \sqrt{|d_K|} L(1, \chi_{d_K}) / (2\pi)$. **PROVED** Splitting the L -series into blocks of length $|d_K|$ and using $\sum_a \chi(a) = 0$ per block gives $|L(1, \chi)| \leq \log|d_K| + 2$; with $w_K \leq 6$ and $\prod_{\ell|f} (1 + 1/\ell) \leq \log f + 1$ the order bound follows. Substituting $|\Delta_{\text{End } E}| \leq 4q$ and checking the ratio to $q/2$ decreases past $q = 2^{21}$ gives the threshold. $\square$

PROVED This does not exclude a **separately constructed** order of much larger discriminant, nor the regime $r \ll q$ — which is precisely why the residual survives.

8.3 Ray classes and their transparency

PROVED For $K = \mathbb{Q}(i)$ and odd r , Milne’s ray-class formula [20] gives $|\text{Cl}_r(\mathbb{Q}(i))| = (r-1)^2/4$ if $r \equiv 1 \pmod{4}$ and $(r^2-1)/4$ if $r \equiv 3 \pmod{4}$, so r does not divide the modulus- r ray class number. **PROVED** At modulus r^2 the r -adic valuation is 2, and the congruence subgroup linearises:

$$(1 + r\mathbb{Z}[i]) / (1 + r^2\mathbb{Z}[i]) \cong (\mathbb{Z}[i]/r\mathbb{Z}[i], +), \quad 1 + rz \mapsto z, \quad (3)$$

because $(1 + rx)(1 + ry) \equiv 1 + r(x + y) \pmod{r^2}$. **PROVED** This yields an order- r target with an immediate logarithm — but $[m]P \mapsto 1 + rm$ is only an abstract homomorphism, since evaluating it from $[m]P$ is the source DLP. The principal units move **lifts** inside one fibre of multiplication-by- r , not the r source points, and full level rescaling reaches the multiples mP through $(\mathbb{Z}/r\mathbb{Z})^\times$, a multiplicative action of order $r-1$ that omits the identity and does not respect the additive source law. Figure 1’s ray cell is excluded on these grounds, sharpened in §9.

9 The evaluator sandwich and the surviving residual

The class-group branch is not closed by any size argument, and the reason is a two-way reduction that we make explicit.

Theorem 13 (evaluator sandwich). **PROVED** Let $\varphi : C \rightarrow G$ be a fixed nonzero homomorphism with all public setup supplied, $h = \varphi(P)$ of order r , so $\varphi(xP) = h^x$. Write $T_{\text{src}}, T_{\text{eval}}, T_{\text{tgt}}$ for source-DLP, evaluation, and target-DLP costs. Then

$$T_{\text{eval}} \leq T_{\text{src}} + O(\log r)T_G, \quad T_{\text{src}} \leq T_{\text{eval}} + T_{\text{tgt}} + (\log r)^{O(1)}. \quad (4)$$

Hence, given a target-DLP oracle, evaluation and source DLP are polynomial-time Turing equivalent; a polynomial evaluator plus an $\exp(o(\log r))$ target DLP is exactly a subexponential source ECDLP reduction.

Proof. Source DLP recovers $x = \log_P Q$ and computes h^x by double-and-add, giving the first bound. Evaluation followed by target DLP computes $Y = \varphi(Q)$ and recovers $x = \log_h Y$, giving the second. The two together are a Turing equivalence. $\square$

PROVED The sandwich is the honest heart of the problem. It shows that target ideal arithmetic and class-number estimates **alone** can never give an unconditional negative answer: once a known order- r target element exists, the only remaining content is the complexity of converting a concrete source point into the corresponding power of that element. A universal impossibility theorem for the residual evaluator must therefore restrict its coordinate/bit/lift interface or prove a genuinely new lower bound for the concrete source family.

9.1 Base classification and the discriminant window

PROVED The possible **bases** of a class target collapse cleanly. Every finite or local base has trivial Picard group, since every invertible module over a local ring is free of rank one and a finite ring is a product of local ones [13]. A pointed global function-field class group is $\text{Pic}^0(C') \cong J_{C'}(\mathbb{F}_q)$ [21], a Jacobian target already covered by §7; a modulus produces a generalised Jacobian, a mixed target also covered. Thus a genuinely distinct class target must be a global number-field order.

PROVED Standard dense global lifts are excluded by Parent’s torsion bound [22]: a rational order- r point over a degree- d number field forces $r \leq 65(3d - 1)(2d)^6$, so $d > (r/12480)^{1/7}$ and the lifted encoding is exponential in n .

The surviving class target must therefore satisfy a two-sided discriminant constraint, from which the window is immediate.

Proposition 14 (discriminant window). **PROVED** If an imaginary-quadratic order of discriminant bit length $B = \log_2 |\Delta|$ contains an order- r class, the class-number bound forces $B \geq 2n - O(\log n)$ with $n = \lceil \log_2 r \rceil$. **CONDITIONAL: ERH + factor-base decomposition** The Hafner–McCurley target route [17] costs $\exp(O(\sqrt{B \log B}))$, which is $\exp(o(n))$ exactly when $B \log B = o(n^2)$. Hence the checked target route requires

$$2n - O(\log n) \leq B = o(n^2 / \log n). \quad (5)$$

PROVED The window is nonempty but strict: a target discriminant must carry roughly twice the source bit length, yet stay below the $n^2 / \log n$ scale or the subexponential target advantage evaporates. Polynomial bit length alone is insufficient; the growth exponent must land in the window.

9.2 Why the ray shortcut is not a shortcut

One might hope to substitute the transparent modulus- r^2 ray target for the opaque ordinary class group. It fails for a sharp reason.

Proposition 15 (ray evaluator equivalence). **PROVED** For the linearisation $\theta(1 + rz) = z \bmod r$ and any nonzero homomorphism $\psi : C \rightarrow U_r$ with $U_r = (1 + r\mathbb{Z}[i]) / (1 + r^2\mathbb{Z}[i])$, pick a nonzero coordinate u_j of $u = \theta(\psi(P))$; on $Q = xP$ the same coordinate of $\theta(\psi(Q))$ is xu_j ,

so one evaluator call and a field inversion return x . Thus a ray evaluator and source DLP are polynomial-time equivalent.

PROVED Because the target logarithm is transparent in the ordinary encoding, any nonzero point evaluator into U_r is **already** a polynomial-time source-DLP algorithm; the ray target has no intermediate subexponential regime. This is why the residual must be an ordinary number-field class group whose target DLP is subexponential rather than polynomial. Two logically separate tasks remain: **construct** a certified exact order- r class in a polynomial-bit order inside the window, and **evaluate** a nonzero homomorphism from finite-field coordinates into it. The sandwich shows solving the first does not supply the second.

10 Empirical validation

Every positive claim about the two known transfers is backed by an end-to-end toy run, and every negative probe by an exhaustive enumeration. Table 2 collects the parameters; the scripts recover fixed seeded logarithms and log the work actually performed.

Table 2: Validation audit. Positive rows recover a fixed known logarithm and confirm the homomorphism; negative rows exhaust a toy family and record the obstruction. Parameters and every recovered value are in the cited CSVs.

Transfer / probe	Parameters	What is measured	Status
Additive (anomalous)	7 curves, $101 \leq p \leq 6421$, $r = p$	$\#E = p$, 5 scalar-homomorphism checks/curve, seeded log recovered	validated
Pairing (MOV/Tate)	7 subgroups, (p, r) from $(43, 11)$ to $(8011, 2003)$, $k = 2$	order + k checks, 4 bilinearity checks/curve, seeded log recovered	validated
Buell reduction	10 reductions, $23 \leq p \leq 59$, $13 \leq r \leq 37$, 218 points	distinct lifted discriminants, points hitting model $\mathcal{D}$	negative
CM class labels	8 $j = 1728$ curves, $13 \leq p \leq 421$, 368 points	distinct CM/annihilator/kernel/Vélu labels per subgroup	negative
Exact-order census	$ \Delta \leq 200000$, primes $3 \leq r \leq 43$	least $ \Delta $ with $r \mid h(\Delta)$, exact-order form	existence only

10.1 The two transfers scale as predicted

EMPIRICAL: Windows 11, Python 3.13.4, 50 timing repeats The median additive-map time fits a work exponent of 1.050508 against lifted group-operation count, and the affine Tate map fits 1.052569 against Miller-line count times field bit length, with all log-residuals stored in `run_transfers_full_20260626_scaling.csv`. Figure 2 shows the fits. **PROVED** The additive evaluator uses double-and-add for the scalar p and so $O(\log p)$ lifted operations, with recorded counts rising from 10 to 18; the Tate map uses an $O(\log r)$ Miller loop, counts rising from 5 to 17. **PROVED** Critically, the toy multiplicative logarithms are solved by BSGS of width $\lceil \sqrt{r} \rceil$, not index calculus, so the timings validate the **map and recovered answer** but do **not** establish the finite-field subexponential asymptotic that requirement (4) invokes in principle — a limitation we mark rather than hide.

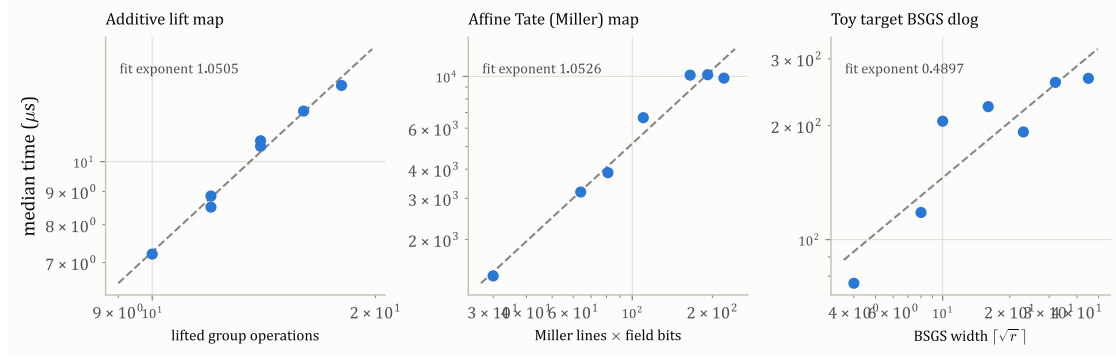


Figure 2: Measured median evaluator time versus the explicit work predictor for the additive lift, the affine Tate map, and the toy target BSGS solver, with the recorded power-law fits (dashed). Fit exponents 1.0505, 1.0526, 0.4897 are read directly from the experiment driver. Source: `run_transfers_full_20260626_scaling.csv`.

10.2 The naive class-group formula does not even land in one group

EMPIRICAL: 10 reductions, 218 nonzero points Applying the explicit Buell point-to-form formula to canonical integer representatives of a finite-field point produces, on almost every point, a **different** lifted discriminant $\mathcal{D} + k_Q p$: complete enumeration found 218 distinct lifted discriminants — one per point — and only two equal to the model discriminant, with every discrepancy divisible by p (`probe_buell_reduction_full_20260710.csv`). **Figure 3** shows this per subgroup. **PROVED** Since binary quadratic forms of different discriminants represent classes of different orders, the outputs do not share a target group; forcing a fixed discriminant means solving the integral curve equation exactly, which returns to the excluded dense-lift branch.

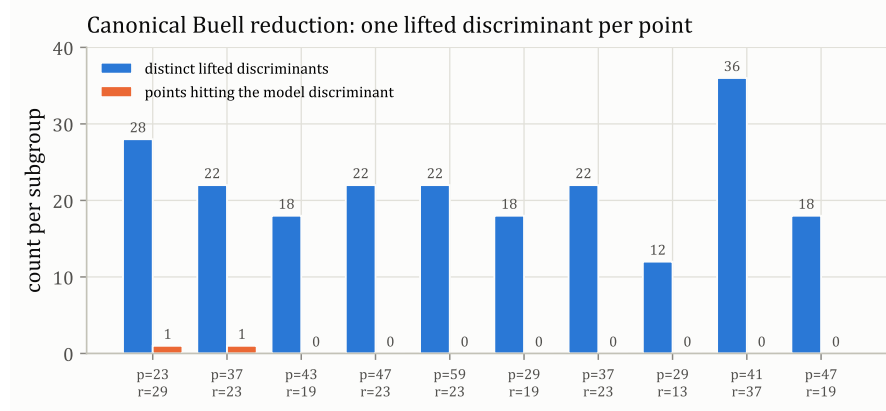


Figure 3: Canonical Buell reduction over ten toy prime-order subgroups: nearly every nonzero point yields its own lifted discriminant (blue), and almost none hit the fixed model discriminant (orange). The map does not land in one class group. Source: `probe_buell_reduction_full_20260710.csv`.

10.3 Small exact-order targets exist, but not uniformly

EMPIRICAL: exhaustive $|\Delta| \leq 200000$, primes $3 \leq r \leq 43$ For every tested prime the least discriminant with $r \mid h(\Delta)$ in fact had $h(\Delta) = r$, with $|\Delta|/r^2 \in [0.684711, 2.555556]$ (`probe_exact_order_targets_full_20260710.csv`); each carries a nonprincipal reduced form of exact order r . **Figure 4** contrasts these succinct census targets with the self-certifying family $\Delta_r = 1 - 4 \cdot 2^r$, which yields a provable exact order- r ideal class $\mathfrak{a} = (2, \alpha)$ but has $\Theta(r)$ discriminant bits and violates the input budget. **PROVED** The census locates each target by

exhaustive class-number search, whose cost is not polynomial in n ; it demonstrates small-target **existence**, not a uniform growing-family constructor. [CITED](#) Prescribed-order existence theorems [23], [24], [25] use n -th-power discriminant families or ineffective thresholds; [EMPIRICAL: bounded primary-source search 2026-07-10](#) no checked theorem supplied a uniform polynomial-bit exact-order constructor.

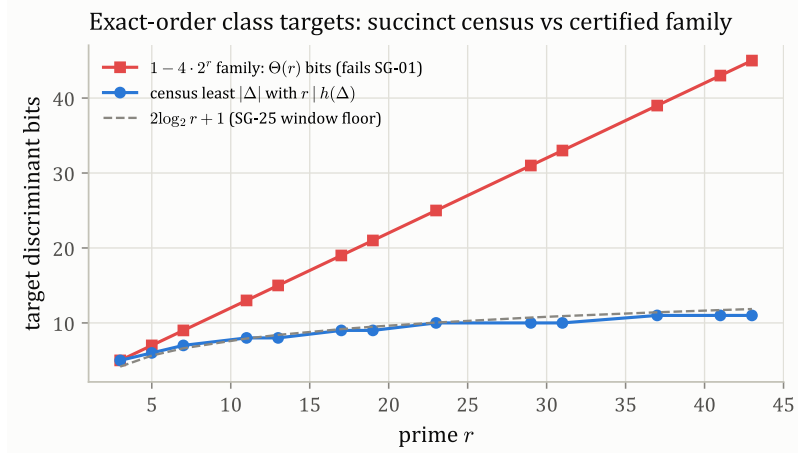


Figure 4: Target discriminant bit length versus prime r . The self-certifying $1 - 4 \cdot 2^r$ family (red) grows linearly in r and fails the input budget; the census least $|\Delta|$ with $r \mid h(\Delta)$ (blue) tracks the window floor $2 \log_2 r + 1$ (dashed) but is found only by exhaustive search. Source: `probe_exact_order_targets_full_20260710.csv`.

11 The restricted classification and its residual

We can now state what is settled.

Restricted classification. Let $C = \langle P \rangle$ have prime order r and consider a transfer family. [PROVED](#) (1) If the source is available only through a random generic encoding with $r^{o(1)}$ queries, no transfer exists. (2) If the target is a smooth proper algebraic group and the evaluator uses fewer than r rational branches, it is one global homomorphism with trivial or elliptic-isogenous image. (3) If the target is affine and the evaluator is piecewise rational with B branches of pole degree $\leq D$, then $\max(1, D)B^2 \geq r/4$, and a rational circuit obeys $d + 2b \geq n - O(\log n)$. (4) Finite/local class targets are trivial and function-field ones are Jacobians; only a global number-field order is genuinely distinct. [CITED](#) The anomalous, pairing, and qualifying Weil-descent transfers occupy the three known exits: a non-rational lift, a high-degree short-program pairing, and the proper/geometric side.

[CONJECTURE](#) The residual class is precise: a structurally new transfer must escape the generic theorem by a representation-specific source operation, and if it targets a class group it must be a **direct cross-characteristic evaluator** taking the ordinary representation of $Q \in E(\mathbb{F}_q)[r]$ to an ideal class of a separately constructed number-field order inside the discriminant window — with no dense global lift supplied as advice. [EMPIRICAL: bounded literature search 2026-07-10](#) Every checked point-to-class construction — Buell [26], Soleng [27], Buell–Call [28], Gillibert [29], Blum et al. [30] — uses characteristic-zero rational or algebraic points, not finite-field source points; this is a

documented search result, not a nonexistence theorem. One explicit family meeting the transfer definition and lying outside all three known mechanisms would refute the negative assessment.

12 Conclusion

The problem asked for a classification of curves admitting a polynomial-time injective homomorphism into an easy-DLP group, together with an explicit decision about Weil descent, validated demonstrations of both known transfers, a common structural description, a candidate table, and a restricted classification. We delivered all of these in their honest form. The two elementary transfers are one character construction (Table 1); Weil descent counts as a known third geometric family; and the candidate table (Figure 1) has a proof, a citation, or an isolated open cell behind every verdict. The structural core is a chain of exclusions — algebraic-group rigidity, generic-source impossibility, rational degree/branch/depth bounds, and proper/mixed rigidity — that together rule out every generic and rational construction and force any nontrivial global image back onto the elliptic side. The empirical layer validates both transfers end to end and confirms each negative probe at toy scale.

What we did **not** do is solve the unrestricted problem, and we are explicit about why. The evaluator sandwich shows the last surviving candidate — a direct cross-characteristic evaluator into an ordinary number-field class group inside the window $2n - O(\log n) \leq B = o(n^2 / \log n)$ — cannot be excluded by class-number size, and no checked construction supplies or forbids it. That single object is the entire distance between our restricted classification and a full one. Naming it exactly, with the models in which it is ruled out and the one in which it is not, is the contribution.

Reproducibility

All computations use toy parameters under the repository’s 60-bit ceiling. The two transfer demonstrations and their timing matrices are produced by `run_transfers.py` (raw and scaling CSVs dated 20260722); the CM/annihilator, ray-class and principal-unit probes by `probe_cm_class_targets.py`; the Buell residue-coordinate experiment by `probe_buell_reduction.py`; the exact-order census by `probe_exact_order_targets.py`; and the finite piecewise-overlap falsification certificate by `audit_rational_tradeoffs.py`. Each writes a seeded CSV named in the captions and figures above. The three figures are regenerated by `figures/P1.5/make.py` directly from those CSVs. Every mathematical claim carries one of the epistemic tags `PROVED`, `CITED`, `EMPIRICAL: range`, `CONDITIONAL: hypothesis`, or `CONJECTURE` exactly as used in the research log; untagged sentences are exposition, not claims. In particular the toy multiplicative targets are solved by BSGS, so they validate the map and answer but not the finite-field subexponential asymptotic, and the literature and census results are bounded searches, not nonexistence or uniformity theorems.

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